Rhulani Osborn Manganyi
Data Analytics
Brightlearn
Exercise number 2
Aggregate functions
Question 01:

01. List all distinct department in the table students

```
19  SELECT * FROM excercises_2.aggregate.students;
20  --- question 1: list all distinct department
21  SELECT DISTINCT department
22  FROM excercises_2.aggregate.students;
23
24
25
```

Add parameter

Table

	department
1	IT
2	HR
3	Finance

02. Get the average age of students per department

```
23  --- question 2: get the average age of student per department
24  SELECT department,
25  AVG(age) AS avg_age
26  FROM excercises_2.aggregate.students
```

Add parameter

Table

	department	1.2 avg_age
1	IT	20.5
2	HR	22
3	Finance	23

03. Show department with more than one student

```sql
--- question 3: show the department with more than one student
SELECT department,
COUNT(student_id) AS student_count
FROM excercises_2.aggregate.students
GROUP BY department
```

Add parameter

Table

	department	student_count
1	IT	2
2	HR	2

04. Get all students whose age is between 21 and 23

```sql
--- question 4: get all students whose age is between 21 and 23
SELECT name,
      age,
      department
FROM excercises_2.aggregate.students
WHERE age BETWEEN 21 AND 23;
```

Add parameter

Table

	name	age	department
1	Bob	22	HR
2	Charlie	21	IT
3	Diana	23	Finance
4	Eve	22	HR

05. List all students in the IT or HR department who are older than 21

```sql
--- question 5: list all students in the IT or HR department who are older than 21
SELECT student_id,
      name,
      age,
      department
FROM excercises_2.aggregate.students
WHERE department IN ('IT', 'HR') AND age > 21;
```

Add parameter

Table

	student_id	name	age	department
1	2	Bob	22	HR
2	5	Eve	22	HR

06. Show total credits per department, only show for departments with more than 5 total credits

```
62  --- question 6: show the total credits per department, only for departments with more than 5 credits
63  SELECT department,
64  SUM(credits) AS total_credits
65  FROM excercises_2.aggregate.courses
66  GROUP BY department
67  HAVING total_credits > 5;
```

Add parameter

Table

	department	total_credits
1	IT	11

07. List all courses that do not have 4 credits

```
68  --- question 7: list all courses that do not have 4 credits
69  SELECT course_id,
70         course_name,
71         credits
72  FROM excercises_2.aggregate.courses
73  WHERE credits != 4;
```

Add parameter

Table

	course_id	course_name	credits
1	101	SQL Basics	3
2	104	Excel	2
3	105	Statistics	3

08. Show the top 3 courses by credits in descending order

```
74  --- question 8: show all the top 3 courses by credits in descending order
75  SELECT course_id,
76         course_name,
77         credits
78  FROM excercises_2.aggregate.courses
79  ORDER BY credits DESC
80  LIMIT 3;
```

Add parameter

Table

	course_id	course_name	credits
1	102	Python	4
2	103	Data Science	4
3	101	SQL Basics	3

09. Get minimum, maximum, and average grade across all enrollments

```sql
--- question 9: Get the maximum, minimum, and average across all enrollments
SELECT MAX(grade) AS max_grade,
       MIN(grade) AS min_grade,
       AVG(grade) AS avg_grade
FROM excercises_2.aggregate.enrollments;
```

Add parameter

Table

	max_grade	min_grade	avg_grade
1	90	78	84.6

10. Count how many enrollment exists per course

```sql
--- question 10: count how many enrollment exists per course
SELECT course_id,
       COUNT(enrollment_id) AS enrollment_count
FROM excercises_2.aggregate.enrollments
GROUP BY course_id;
```

Add parameter

Table

	course_id	enrollment_count
1	101	1
2	102	1
3	103	1
4	104	1
5	105	1

11. Find the salary and total bonus per department

```
122  --- question 11: find total salary and total bonus per department
123  SELECT department,
124         SUM(salary) AS total_salary,
125         SUM(bonus) AS total_bonus
126  FROM excercises_2.aggregate.salaries
127  GROUP BY department;
128
129
```

Add parameter

Table

	department	total_salary	total_bonus
1	IT	122000	10500
2	HR	109000	7500
3	Finance	70000	6000

12. Show department where average salary is above 55000

```
128  --- question 12: show department where average salary is above 55000
129  SELECT department,
130         AVG(salary) AS avg_salary
131  FROM excercises_2.aggregate.salaries
132  GROUP BY department
133  HAVING avg_salary > 55000;
134
135
136
```

Add parameter

Table

	department	avg_salary
1	IT	61000
2	Finance	70000

13. List employees whose salary plus bonus is greater than 60000

```sql
--- question 13: list all employees whose total salary plus bonus is greater than 60000
SELECT employee_id,
       name,
       salary,
       bonus,
       (salary + bonus) AS total_compensation
FROM excercises_2.aggregate.salaries
WHERE (salary + bonus) > 60000;
```

Add parameter

Table

	employee_id	name	salary	bonus	total_compensation
1	1	Tom	60000	5000	65000
2	3	Spike	70000	6000	76000
3	4	Tyke	62000	5500	67500

14. Show total and average budget per department. Only include departments with average budget above 70000

```sql
--- question 14: show total and averagr budget per department. only include department with average budget above 70000
SELECT department,
SUM(budget) AS total_budget,
AVG(budget) AS avg_budget
FROM excercises_2.aggregate.projects
GROUP BY department
HAVING avg_budget > 70000;
```

Add parameter

Table

	department	total_budget	avg_budget
1	IT	270000	135000
2	Finance	80000	80000

15. List all projects with budgets between 50000 and 120000, excluding the marketing department.

```sql
--- question 15: list all projects with budgets between 50000 and 120000, excluding the marketing department
SELECT project_id,
       project_name,
       department,
       budget
FROM excercises_2.aggregate.projects
WHERE budget BETWEEN 50000 AND 120000 AND department != 'Marketing';
```

Add parameter

Table

	project_id	project_name	department	budget
1	1	AI App	IT	120000
2	2	Payroll System	Finance	80000
3	5	HR Portal	HR	50000